
Smart Disaster Early-Warning and Emergency Coordination System for Sri Lanka**Introduction**

Sri Lanka faces recurring natural hazards including monsoon flooding, landslides in the central highlands, cyclones along the coast, and occasional drought. When these events strike, the effectiveness of the response often depends less on the severity of the hazard itself and more on how quickly warnings reach the public, and on how efficiently relief resources are coordinated across multiple agencies. Those resources include rescue teams, shelters, medical supplies, and volunteers.

Today, this coordination is carried out largely through phone calls, radio broadcasts, WhatsApp groups, and paper-based situation reports exchanged between the Disaster Management Centre (DMC), district secretariats, the police, the armed forces, and non-governmental organisations. This fragmented approach leads to duplicated efforts, delayed evacuations, and relief supplies that do not reach the areas of greatest need. Several countries have adopted integrated digital platforms to address exactly this problem, such as Japan's J-Alert early-warning network and the United States' FEMA disaster management systems.

Objective

In this case study, your team is bidding to design a unified digital platform for the Disaster Management Centre of Sri Lanka. The goal is to build a system that issues timely, location-specific hazard warnings to the public. It should also coordinate the response effort across government agencies, first responders, relief camps, and volunteers, for hazards of differing scale and type.

- **First Phase:** Develop a design for your software that offers full functionality to address the complexity of multi-agency disaster response, as well as high usability for both civilians and field responders operating under stressful, time-critical conditions.
- **Second Phase:** Develop, test, and evaluate a prototype for your software platform, demonstrating both the usability features and the flexibility of the system to be configured for different hazard types and different districts.

Requirements

In the first phase, you are required to demonstrate your understanding of the domain by creating a software functional design (using UML) and user interaction designs for the software.

Case Study Details

The proposed system is built around a citizen mobile application and web portal linked to a central hazard-monitoring dashboard used by DMC officers. Citizens register with their national identity number, home address, and district, which is used to determine which hazard alerts are relevant to them. Alerts can also be broadcasted more broadly by district or by river basin, depending on the nature of the hazard. Citizens receive push notifications, SMS messages, and audible alerts even when the app is running in the background. This matters because network connectivity during a disaster cannot be guaranteed to be reliable.

Hazard Reporting and Verification

In addition to alerts issued by the DMC, ordinary citizens and community-level disaster volunteers should be able to submit ground reports, for example a rising river level, a blocked road, or a landslide crack, along with a photo and GPS location.

These citizen reports need to be reviewed and verified by a duty officer before they influence any official warning level, since unverified reports could cause unnecessary panic or, conversely, be dismissed as false alarms if not handled carefully.

Resource and Shelter Coordination

When an official warning is escalated, the system must help district officers coordinate the response. This includes registering available emergency shelters and tracking their current occupancy against capacity. It also includes dispatching rescue teams and recording their real-time status, and logging relief supplies such as food, water, and medicine as they are distributed.

Some resources are owned by government bodies, others by the armed forces, and others by NGOs and private donors. The system must be able to represent resources owned and controlled by different organisations while still giving DMC officers a combined operational picture.

System Flexibility

The system must be designed to handle hazards that vary enormously in speed and scale, from a flash flood that develops in hours to a slow-onset drought that develops over months. It must also be usable during power and network outages in the affected area itself, even though the central dashboard used

by DMC officers may be located elsewhere with normal connectivity. The design should account for graceful degradation, offline data capture, and delayed synchronisation where appropriate.

Data Analysis and Reporting

After an event, DMC officials and donor organisations need to analyse what happened in order to improve future response and to report on how relief resources were used. The system should generate statistical reports on the timeline of alerts issued, the number of citizens reached, shelter occupancy over time, and resource distribution by district. That way, officials can evaluate response effectiveness and plan for future events.

Existing Systems

Different countries have taken different approaches to disaster early-warning and coordination, shaped by their own hazards, infrastructure, and administrative structures. The following resources provide insight into various disaster management systems around the world:

- [Sri Lanka Disaster Aid Management Platform \(SL Relief\)](#)
- [Japan's J-Alert National Warning System](#)
- [USA's FEMA Disaster Management Systems](#)
- [EU's Copernicus Emergency Management Service](#)

Important Note

The information in this document is typical of what you might expect from a client in the industry when developing software systems: possibly incomplete, ambiguous, and vague. You are encouraged to discuss any queries with the module delivery team during tutorials to clarify any aspects of the system design and implementation.

Remember, your clients will rarely be computer-savvy. It is your job as practicing software engineers to translate their requirements into formal designs and ultimately a functional software package. If the clients could do this themselves, they wouldn't need our expertise.